ClipSAT : Cambridge AS Level Math 9709

Unit 0.01: AS-Level 9709 Paper 1/2/4/5 Diagnostic Readiness Name: _____

Q1. The diagram shows a right-angled triangle ABC, with the right angle at D. The altitude from D to the hypotenuse BC is drawn. Given that BD = 4 cm and DC = 9 cm, find the length of the altitude AD.

[ ] A: 6 cm
[ ] B: 5 cm
[ ] C: 7 cm
[ ] D: 8 cm

Q2. A fair six-sided die is rolled 4 times. Let X be the number of times a 6 is obtained. The possible values for X are 0, 1, 2, 3, and 4. The probability of obtaining a 6 on any given roll is p = frac{1}{6} . The number of trials is n=4 . This scenario can be modelled by a binomial distribution, denoted as X sim Bn, p . Which of the following represents the correct parameters for this binomial distribution?

[ ] A: B4, 1/6
[ ] B: B6, 1/4
[ ] C: B1, 4/6
[ ] D: B4, 4/6

Q3. The diagram shows part of the curve with equation y = x^2 - 4x + 5 and the point P3, 2 lying on the curve. Find the gradient of the tangent to the curve at the point P .

[ ] A: 2
[ ] B: 6
[ ] C: -4
[ ] D: dfrac{2}{3}

Q4. The diagram shows a triangle ABC with coordinates A2, 1, B8, 1 and C5, 5. Find the length of the side AC.

[ ] A: 5
[ ] B: sqrt{26}
[ ] C: 6
[ ] D: 7

Q5. The diagram shows a right-angled triangle ABC, with the right angle at D. The altitude from D to the hypotenuse BC is drawn. Given that BD = 4 cm and DC = 9 cm, find the length of the altitude AD.

[ ] A: 6 cm
[ ] B: 5 cm
[ ] C: 7 cm
[ ] D: 8 cm

Q6. A fair six-sided die is rolled 4 times. Let X be the number of times a 6 is obtained. The possible values for X are 0, 1, 2, 3, and 4. The probability of obtaining a 6 on any given roll is p = frac{1}{6} . The number of trials is n=4 . This scenario can be modelled by a binomial distribution, denoted as X sim Bn, p . Which of the following represents the correct parameters for this binomial distribution?

[ ] A: B4, 1/6
[ ] B: B6, 1/4
[ ] C: B1, 4/6
[ ] D: B4, 4/6

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Q7. The diagram shows part of the curve with equation y = x^2 - 4x + 5 and the point P3, 2 lying on the curve. Find the gradient of the tangent to the curve at the point P .

[ ] A: 2

[ ] B: 6

[ ] C: -4

[ ] D: dfrac{2}{3}

Q8. The diagram shows a triangle ABC with coordinates A2, 1, B8, 1 and C5, 5. Find the length of the side AC.

[ ] A: 5

[ ] B: sqrt{26}

[ ] C: 6

[ ] D: 7